

Francesco Schiavone

Curriculum vitae

Last updated July 31, 2026

Work experience

INFN Research Fellow INFN Sezione di Bari Research topic: <i>Study of cosmic radiation with Fermi-LAT</i>	June 2026 – present Bari, Italy
Peer tutor for the General Physics I and II courses Università degli Studi di Bari “Aldo Moro”	October 2025 – April 2026 Bari, Italy
Peer tutor for the General Physics II course Università degli Studi di Bari “Aldo Moro”	March – December 2024 Bari, Italy

Education

PhD in Astroparticle Physics Università degli Studi di Bari “Aldo Moro” Thesis title: <i>Present and future constraints on axion-like particles from blazars: from current gamma-ray telescopes to the Cherenkov Telescope Array Observatory</i> Supervisors: Prof. Francesco Giordano, Dr. Leonardo Di Venere	February 2023 – May 2026 Bari, Italy
Master’s degree in Theoretical Physics Università degli Studi di Bari “Aldo Moro” Thesis title: <i>Fluxes and signatures of gravitons and axions from the evaporation of primordial black holes</i> Thesis supervisors: Prof. Alessandro Mirizzi, Prof. Daniele Montanino Grade: 110/110 <i>cum laude</i>	September 2018 – April 2021 Bari, Italy
Bachelor’s degree in Physics Università degli Studi di Bari “Aldo Moro” Thesis title: <i>Conservation of parity</i> (original: <i>La conservazione della parità</i>) Thesis supervisor: Prof. Domenico Di Bari Grade: 110/110 <i>cum laude</i>	September 2015 – July 2018 Bari, Italy
High school diploma Liceo Scientifico Statale “Arcangelo Scacchi” Grade: 100/100 <i>cum laude</i>	September 2010 – July 2015 Bari, Italy

Affiliations

MAGIC Collaboration Member	October 2024 – present
COST Action CA21106 Theme: <i>COSMIC WISPerS in the Dark Universe: Theory, astrophysics and experiments</i> Member of Working Groups 3 (WISPs in Astrophysics) and 4 (Direct WISPs searches)	June 2023 – present
Cherenkov Telescope Array Observatory (CTAO) Consortium Member	March 2023 – present

CTAO Large-Sized Telescope (CTAO-LST) Collaboration
Member

March 2023 – present

Istituto Nazionale di Fisica Nucleare (INFN) – Sezione di Bari
Member

March 2023 – present

Research activity

My research focuses on the study of high energy and very high energy gamma rays from astrophysical sources. In particular, my main interests include active galactic nuclei studies and indirect searches of dark matter and axion-like particles (ALPs) with the Cherenkov Telescope Array Observatory (CTAO) and other ground-based gamma-ray telescopes. My work on ALPs started during my master's degree in Theoretical Physics, with a particle cosmology thesis discussing the possible observable effects of a population of primordial black holes radiating spin-0 and spin-2 bosons before the epoch of Big Bang nucleosynthesis. The main results of this work were later the subject of a journal publication [24].

My growing interest in data analysis and experimental techniques led me to start a PhD within the experimental astroparticle physics group at the University of Bari and INFN, with a scholarship funded by the Italian PNRR CTA+ project. During my PhD I have worked on the simulation and statistical analysis of blazar observations with CTAO, to estimate the sensitivity of the observatory to ALP signals at TeV energies while exploring the possibility of using new analysis methods based on machine learning classifiers. The results of this work are collected in a published paper [1]. I have also performed a long-term analysis of the observations of four blazars by the MAGIC telescopes over a period of 10 years, aimed at constraining the presence of ALP signatures by stacking data relative to both low and flaring states in combination with 4 years of LST-1 data. Furthermore, I contributed to a detectability study of flat-spectrum radio quasars (FSRQs) with CTAO, performing simulations and statistical analysis to estimate CTAO's capability of distinguishing between different spectral shapes. Both these works will be the subjects of two publications, currently in preparation. More recently, I have been involved in the production of ALP models for the upcoming CTAO Open Science Data Challenge.

I have experience with low- and high-level gamma-ray analysis software such as the MAGIC Analysis and Reconstruction Software (MARS), lstchain and Gammapy. As a member of the CTAO, LST and MAGIC collaborations I am frequently involved in technical activities related to telescope operation, such as data-taking shifts and online data checks.

Activity in international collaborations

LST Cherenkov camera assembly for CTAO South
University of Catania (IT)

25 – 28 May 2026

MAGIC Operator
Observatorio del Roque de los Muchachos, La Palma (ES)

5 – 29 March 2026 (P287)

LST-1 Deputy Shift Leader
Observatorio del Roque de los Muchachos, La Palma (ES)

15 July – 4 August 2025 (P079)

LST-1 Flare Advocate
Remote

18 March – 7 April 2025 (P075)

LST-1 Deputy Flare Advocate
Remote

19 November – 11 December 2024 (P071)

LST-1 Operator
Observatorio del Roque de los Muchachos, La Palma (ES)

27 February – 19 March 2024 (P062)

Lectures and seminars

Seminar for the students of the Multimessenger Astrophysics course 10 November 2025

Università degli Studi di Bari “Aldo Moro”

Title: *Probing dark matter and fundamental physics with Cherenkov telescopes*

Seminar for the students of the High Energy Astrophysics course 26 March 2024

Università degli Studi di Bari “Aldo Moro”

Title: *Basics of optical telescopes*

Invited seminar 12 December 2021

University of Zaragoza (ES)

Title: *Axion-like particles from primordial black holes shining through the Universe*

Virtual Axion Institute 19 July 2021

Online ([YouTube link](#))

Title: *Axion-like particles from PBHs shining through the Universe*

Publications

I am co-author of 25 papers and proceedings, with a h -index of 7 according to [INSPIREHEP](#). The full list of publications is available on [ORCID](#) and reported below:

- [1] F. Schiavone, L. Di Venere and F. Giordano. “Projected sensitivity of CTAO to axion-like particles from blazars with a machine learning approach”. *Eur. Phys. J. C* **86**, 639 (2026). doi: [10.1140/epjc/s10052-026-15910-x](#). arXiv: [2512.19259](#) [[astro-ph](#)].
- [2] S. Abe *et al.* “Chasing Gamma-Ray Signals from Binary Neutron Star Coalescences with the Cherenkov Telescope Array: Prospects and Observing Strategies”. *Astrophys. J.* **1004** (2026), 46. doi: [10.3847/1538-4357/ae658c](#). arXiv: [2604.08748](#) [[astro-ph](#)].
- [3] K. Abe *et al.* “Extreme Blazars Observed with MAGIC: Second Catalog Release”. *Astrophys. J.* **1002** (2026), 138. doi: [10.3847/1538-4357/ae5a8a](#). arXiv: [2604.05576](#) [[astro-ph](#)].
- [4] K. Abe *et al.* “Long-term study of the gamma-ray emission of Cygnus X-3 with MAGIC and Fermi-LAT”. *JHEAP* **53** (2026), 100630. doi: [10.1016/j.jheap.2026.100630](#). arXiv: [2604.27671](#) [[astro-ph](#)].
- [5] K. Abe *et al.* “Very-high-energy gamma-ray intranight variability from BL Lacertae during the extreme flaring state of 2022”. *Astron. Astrophys.* **710** (2026), A41. doi: [10.1051/0004-6361/202557924](#). arXiv: [2603.19889](#) [[astro-ph](#)].
- [6] K. Abe *et al.* “CTAO LST-1 observations of magnetar SGR 1935+2154: Deep limits on sub-second bursts and persistent tera-electronvolt emission”. *Astron. Astrophys.* **706** (2025), A25. doi: [10.1051/0004-6361/202555431](#). arXiv: [2512.05634](#) [[astro-ph](#)].
- [7] F. Schiavone for the CTAO Consortium. “Dark matter and fundamental physics with the Cherenkov Telescope Array Observatory”. *Il Nuovo Cimento* **49 C**, 7 (2026). doi: [10.1393/ncc/i2026-26007-6](#). arXiv: [2509.24523](#) [[astro-ph.HE](#)].
- [8] F. Schiavone, L. Di Venere, F. Giordano for the CTAO Consortium. “A machine learning approach to axion-like particle searches in CTAO observations of blazars”. *Proceedings of 39th International Cosmic Ray Conference — PoS(ICRC2025)* **501**, 834 (2025). doi: [10.22323/1.501.0834](#). arXiv: [2509.24543](#) [[astro-ph](#)].
- [9] L. Nikolić *et al.* on behalf of the CTAO-LST Project and the MAGIC Collaboration. “Recent observations of PKS 2155-304 with MAGIC and LST-1 in a multi-wavelength context”. *Proceedings of 39th International Cosmic Ray Conference — PoS(ICRC2025)* **501**, 783 (2025). doi: [10.22323/1.501.0783](#). arXiv: [2510.04803](#) [[astro-ph](#)].

- [10] K. Abe *et al.* “Prospects for dark matter observations in dwarf spheroidal galaxies with the Cherenkov Telescope Array Observatory”. *Mon. Not. Roy. Astron. Soc.* **544**, 3 (2025), p. 2946–2986. doi: [10.1093/mnras/staf1798](https://doi.org/10.1093/mnras/staf1798). arXiv: [2508.19120](https://arxiv.org/abs/2508.19120) [astro-ph.HE].
- [11] K. Abe *et al.* “VHE γ -ray observations of bright BL Lacs with the Large-Sized Telescope prototype (LST-1) of the CTAO”. *Mon. Not. Roy. Astron. Soc.* **544**, 1 (2025), p. 669–686. doi: [10.1093/mnras/staf1728](https://doi.org/10.1093/mnras/staf1728). arXiv: [2510.03702](https://arxiv.org/abs/2510.03702) [astro-ph.HE].
- [12] K. Abe *et al.* “Constraining the TeV gamma-ray emission of SN 2024bch, a possible type II-L from a red supergiant progenitor. Multiwavelength observations and analysis of the progenitor”. *Astron. Astrophys.* **702** (2025), A125. doi: [10.1051/0004-6361/202554721](https://doi.org/10.1051/0004-6361/202554721). arXiv: [2508.20048](https://arxiv.org/abs/2508.20048) [astro-ph].
- [13] K. Abe *et al.* “Galactic transient sources with the Cherenkov Telescope Array Observatory”. *Mon. Not. Roy. Astron. Soc.* **540**, 1 (2025), p. 205–238. doi: [10.1093/mnras/staf655](https://doi.org/10.1093/mnras/staf655). arXiv: [2405.04469](https://arxiv.org/abs/2405.04469) [astro-ph.HE].
- [14] K. Abe *et al.* “GRB 221009A: Observations with LST-1 of CTAO and Implications for Structured Jets in Long Gamma-Ray Bursts”. *Astrophys. J. Lett.* **988** (2025), L42. doi: [10.3847/2041-8213/ade4cf](https://doi.org/10.3847/2041-8213/ade4cf). arXiv: [2507.03077](https://arxiv.org/abs/2507.03077) [astro-ph].
- [15] F. Schiavone, L. Di Venere, F. Giordano for the CTAO Consortium. “CTAO sensitivity to axion-like particles”. *EPJ Web of Conferences* **319**, 13001 (2025). doi: [10.1051/epjconf/202531913001](https://doi.org/10.1051/epjconf/202531913001).
- [16] A. Cifuentes *et al.* “Performance of MAGIC stellar intensity interferometer and expansion to MAGIC + CTAO-LST1 stellar intensity interferometer”. *SPIE Proceedings Volume 13095, Optical and Infrared Interferometry and Imaging IX* 1309527 (2024). doi: [10.1117/12.3016905](https://doi.org/10.1117/12.3016905). arXiv: [2506.04994](https://arxiv.org/abs/2506.04994) [astro-ph].
- [17] K. Abe *et al.* “Detection of the Geminga pulsar at energies down to 20 GeV with the LST-1 of CTAO”. *Astron. Astrophys.* **698** (2024), A283. doi: [10.1051/0004-6361/202554350](https://doi.org/10.1051/0004-6361/202554350). arXiv: [2505.21632](https://arxiv.org/abs/2505.21632) [astro-ph].
- [18] K. Abe *et al.* “Detection of RS Oph with LST-1 and modelling of its HE/VHE gamma-ray emission”. *Astron. Astrophys.* **695** (2024), A152. doi: [10.1051/0004-6361/202452447](https://doi.org/10.1051/0004-6361/202452447). arXiv: [2503.13283](https://arxiv.org/abs/2503.13283) [astro-ph].
- [19] K. Abe *et al.* “A new method of reconstructing images of gamma-ray telescopes applied to the LST-1 of CTAO”. *Astron. Astrophys.* **691** (2024), A328. doi: [10.1051/0004-6361/202450889](https://doi.org/10.1051/0004-6361/202450889). arXiv: [2410.16042](https://arxiv.org/abs/2410.16042) [astro-ph].
- [20] K. Abe *et al.* “A detailed study of the very high-energy Crab pulsar emission with the LST-1”. *Astron. Astrophys.* **690** (2024), A167. doi: [10.1051/0004-6361/202450059](https://doi.org/10.1051/0004-6361/202450059). arXiv: [2407.02343](https://arxiv.org/abs/2407.02343) [astro-ph].
- [21] K. Abe *et al.* on behalf of the CTA-LST Project. “GRB 221009A observations with LST-1 at VHE gamma rays”. *Proceedings of High Energy Phenomena in Relativistic Outflows VIII — PoS(HEPROVIII)* **461**, 058 (2024). doi: [10.22323/1.461.0058](https://doi.org/10.22323/1.461.0058).
- [22] S. Abe *et al.* “Dark matter line searches with the Cherenkov Telescope Array”. *JCAP* **07** (2024), p. 047. doi: [10.1088/1475-7516/2024/07/047](https://doi.org/10.1088/1475-7516/2024/07/047). arXiv: [2403.04857](https://arxiv.org/abs/2403.04857) [hep-ph].
- [23] S. Abe *et al.* “Prospects for a survey of the Galactic plane with the Cherenkov Telescope Array”. *JCAP* **10** (2024), p. 081. doi: [10.1088/1475-7516/2024/10/081](https://doi.org/10.1088/1475-7516/2024/10/081). arXiv: [2310.02828](https://arxiv.org/abs/2310.02828) [astro-ph].
- [24] F. Schiavone, A. Mirizzi, D. Montanino, F. Capozzi. “Axion-like particles from primordial black holes shining through the Universe”. *JCAP* **08** (2021), p. 063. doi: [10.1088/1475-7516/2021/08/063](https://doi.org/10.1088/1475-7516/2021/08/063). arXiv: [2107.03420](https://arxiv.org/abs/2107.03420) [hep-ph].
- [25] S. Longo, G. Micca Longo, E. Pavone, F. Schiavone. “Deterministic models of minority neutral particle kinetics close to a catalytic surface, based on the formalism of radiative transfer”. *Plasma Sources Sci. Technol.* **28** (2019) p. 125008. doi: [10.1088/1361-6595/ab5150](https://doi.org/10.1088/1361-6595/ab5150)

I have additionally contributed to the following past publications:

- [1] D. Paneque et al. on behalf of the MAGIC Collaboration. “Gamma-ray upper observations of SN 2026sqf with MAGIC”. [AstroNote 2026-222](#).
- [2] D. Paneque et al. on behalf of the MAGIC and LST Collaborations. “Gamma-ray upper limits the on the rising phase of TDE 2025aarm from observations by the MAGIC and LST-1 telescopes”. [AstroNote 2025-328](#).

Contributions to conferences and workshops

MAGIC/LST General Meeting 9 June 2026

University of Padova (IT)

Talk: *ALP constraints from MAGIC and LST-1 blazars*

2nd LST+MAGIC F2F Science Meeting 20 November 2025

Instituto de Astrofísica de Andalucía (IAA), Granada (ES)

Talk: *Long-term blazar analysis for axion-like particle searches with MAGIC and LST-1*

TeVPA 2025 6 November 2025

Fundación ADEIT, Valencia (ES)

Talk: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*

20th Patras Workshop on Axions, WIMPs and WISPs 22–26 September 2025

IACTEC, La Laguna, Tenerife (ES)

Talk/poster: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*

3rd General Meeting of COST Action COSMIC WISPerS (CA21106) 9 – 12 September 2025

University of Sofia (BG)

Invited talk: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*

Extragalactic jets at all scales: a Cretan view 25 – 29 August 2025

Aquila Atlantis Hotel, Heraklion (GR)

Poster/talk: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*

39th International Cosmic Ray Conference (ICRC 2025) 14 – 24 July 2025

International Conference Centre (CICG), Geneva (CH)

Poster: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach* (first author, presented by L. Di Venere)

MAGIC General Meeting 10 June 2025

Complesso San Francesco, Siena (IT)

Talk: *Long-term blazar analysis for axion-like particle searches with MAGIC*

2nd VHEgam Meeting 26 May 2025

Centro Polifunzionale Studenti, Bari (IT)

Talk: *Sensitivity of CTAO to axion-like particles from blazars: a machine learning approach*

Incontri di Fisica delle Alte Energie (IFAE) 2025 10 April 2025

Manifattura Tabacchi, Cagliari (IT)

Talk: *Materia oscura e fisica fondamentale con il Cherenkov Telescope Array Observatory* (in Italian)

V Gravi-Gamma-Nu Workshop 9 – 11 October 2024

Centro Polifunzionale Studenti, Bari (IT)

Poster: *CTAO sensitivity to axion-like particles*

RICAP 2024

23 – 27 September 2024

Hotel Villa Tuscolana, Frascati (IT)

Poster: *CTAO sensitivity to axion-like particles***1st CTAO Summer School**

21 June 2024

Centro Residenziale Universitario, Bertinoro (IT)

Talk: *CTAO sensitivity to axion-like particles***ISAPP23 “Exploring the Dark Universe From Particles to Galaxies”**

13 – 19 November 2023

De Krim, Texel (NL)

Poster: *LHAASO J2108+5157 as an extragalactic source: constraints on axion-like particles from LST-1 data***IV Gravi-Gamma-Nu Workshop**

4 October 2023

Gran Sasso Science Institute, L'Aquila (IT)

Talk: *Astrophysical clues of axion-like particles***IBS-ICTP Workshop on Axion-Like Particles**

29 October 2021

Online ([YouTube link](#))Talk: *Axion-like particles from PBHs shining through the Universe***16th Patras Workshop on Axions, WIMPs and WISPs**

18 June 2021

Online

Talk: *Probing axion-like particle radiation from primordial black holes***PPC 2021: XIV International Workshop on Interconnections between Particle Physics and Cosmology**

19 May 2021

University of Oklahoma (online)

Talk: *Probing axion-like particle radiation from primordial black holes*

Organization of conferences and workshops

*As part of Local Organizing Committee:***2nd VHEgam Meeting**

26 May 2025

Centro Polifunzionale Studenti, Bari (IT)

Graduate schools

1st CTAO Summer School

16 – 29 June 2024

Centro Residenziale Universitario, Bertinoro (IT)

Hacienda San Jorge and Observatorio del Roque de Los Muchachos, La Palma (ES)

2nd LST Analysis School

5 – 9 February 2024

Online

ISAPP23 “Exploring the Dark Universe From Particles to Galaxies”

13 – 19 November 2023

De Krim, Texel (NL)

1st Training School of COST Action COSMIC WISPs (CA21106)

11 – 14 September 2023

Università del Salento, Lecce (IT)

Tre pubblici, tre linguaggi: una comunicazione efficace

13 – 15 June 2023

Science communication workshop

Università degli Studi di Bari “Aldo Moro”

Language skills

Italian (native), English (fluent), Spanish (basic).

IELTS Academic English certification corresponding to C1 CEFR level (band score 8.0/9.0).

Computer skills

General: Familiar with Windows and Linux operating systems. Some experience with Linux-based HPC clusters (ReCaS, PIC) and job scheduling (HTCondor). Some experience with git-based platforms (GitHub, GitLab).

Programming: Familiar with Python, especially data-handling libraries (Numpy, Matplotlib, Pandas) and astrophysical data analysis software (Astropy, Gammapy). Some experience with machine learning packages (scikit-learn, Keras, XGBoost). Basic knowledge of C, C++, FORTRAN, MATLAB and Mathematica, including the ROOT and GEANT4 packages.

Typesetting software: Proficient in the use of LaTeX for papers, books and presentations. Familiar with Microsoft Office software (Word, PowerPoint, Excel).

Multimedia: Familiar with audio, photo and video editing software.

Outreach activities

Together with my research group, I am involved in the organization of several outreach events. These include yearly events such as the European Researchers' Night and International Cosmic Day, as well as sporadic presentations dedicated to the general public and schools. A selection of the most recent activities I have contributed to is found below:

INFN-OCRA Cosmic Ray Cube (CRC) 2025 – present
Local contact for the CRC detector hosted in Bari.

STEM PhD Day at Bari University 5 November 2024
Poster presentation.

Futuro Remoto science festival 18 – 20 October 2024
Città della scienza, Napoli (IT)
Interactive presentations of CTAO and gamma-ray astrophysics dedicated to high school students and general public.

INFN LAB2GO 2024 – present
Tutoring high school students in writing and presenting laboratory reports.

International Cosmic Day (ICD) 2023 – present
Helping in the organization and tutoring of high school students during several editions of the ICD in Bari.

LaTeX and Python courses for Physics students 2019 – present
Organized yearly by the AISF (Italian association of Physics students) Local Committee in Bari, often in collaboration with other Committees, both online and in-person.

European Researcher's Night (ERN)/SHARPER Night in Bari

2018 – present

Presentation of my research group's activities and of simple physics experiments to the general public.

Science photography

I am an amateur photographer, with several of my photographs depicting scientific work and events being used in newspapers, press releases, official social media channels and websites. My photographs of MAGIC and LST-1 received 2nd place in the General category and 6th place in the Astronomy category, respectively, at the Italian national final of the [2025 Wiki Science Competition](#). Another photo of mine is currently featured as the background of the [official LST Collaboration website](#). Some of my science-themed photographs can be found on [my Instagram profile](#).